

Assignment-4 Analysis of Algorithm

Zunaira
Abdul Aziz
BSSC23058-A

Q1

Weight Capacity = 8 kg.

item	1	2	3	4
Profit	12	20	30	40
Weight	2	3	4	5

			Capacity									
	Profit	weight	item	0	1	2	3	4	5	6	7	8
(Empty)	0	0	0	0	0	0	0	0	0	0	0	0
	12	2	1	0	0	12	12	12	12	12	12	12
	20	3	2	0	0	12	20	20	32	32	32	32
	30	4	3	0	0	12	20	30	32	42	50	50
	40	5	4	0	0	12	20	30	40	42	52	60

Item 1 selected $12, 2 \Rightarrow \text{At } 2\text{kg} = 12$

Item 2 selected $12, 2 \Rightarrow \text{At } 2\text{kg} = 12 \mid 20, 3 \Rightarrow \text{At } 3\text{kg} = 20$

$12+20, 2+3 \Rightarrow \text{At } 5\text{kg} = 32$

Item 3 selected $12, 2 \Rightarrow 2\text{kg} = 12 \mid 20, 3 \Rightarrow \text{At } 3\text{kg} = 20 \mid 30, 4 \Rightarrow 4\text{kg} = 30$

$12+20, 2+3 \Rightarrow \text{At } 5\text{kg} = 32 \mid 12+30, 2+4 \Rightarrow \text{At } 6\text{kg} = 42$

$30+20, 4+3 \Rightarrow \text{At } 7\text{kg} = 50$

Item 4 selected $12, 2 \Rightarrow \text{At } 2\text{kg} = 12 \mid 20, 5 \Rightarrow \text{At } 3\text{kg} = 20 \mid 30, 4 \Rightarrow 4\text{kg} = 30$

$40, 5 \Rightarrow \text{At } 5\text{kg} = 40 \mid 12+20, 2+3 \Rightarrow \text{At } 5\text{kg} = 32 \mid 12+30, 2+4 \Rightarrow \text{At } 6\text{kg} = 42$

$40+30, 5+4 \Rightarrow \text{At } 9\text{kg} = 70$
 $\times$ Reject 9kg $\mid 40+12, 5+2 \Rightarrow 7\text{kg} = 52 \mid 40+20, 5+3 \Rightarrow 8\text{kg} = 60$

Items chosen

I_1	I_2	I_3	I_4
12	$V[2][3] = 20$ weight = 3 capacity = 3 - 3 = 0	$V[3][3] = 30$ weight = 4 X	$V[4][8] = 60$ Capacity = 8 - 5 = 3 Profit = 40
0	1	0	1

$$\begin{aligned} \text{Maximum profit} &= 20 \times 1 + 12 \times 0 + 30 \times 0 + 40 \times 1 \\ &= 20 + 0 + 0 + 40 \\ &= 60 \end{aligned}$$

Item chosen = I_2 & I_4

Time complexity = $O(NW)$

N = no of items, w = Max capacity.

Space complexity = $O(NW)$

Q2

capacity, w
weight = 10

	A	B	C	D
items	1	2	3	4
weight	2	5	6	7
profit	10	25	30	35

$$V[i][w] = \begin{cases} 0 & \text{if } i = 0 \text{ or } w = 0 \\ V[i-1][w] & \text{if } w_i > w \\ \max\{V[i-1][w], V[i-1][w-w_i] + p_i\} & \text{if } w_i \leq w \end{cases}$$

Capacity													
Profit	weight	Item	0	1	2	3	4	5	6	7	8	9	10
0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	2	1	0	0	10	10	10	10	10	10	10	10	10
25	5	2	0	0	10	10	10	25	25	35	35	35	35
30	6	3	0	0	10	10	10	25	30	35	40	40	40
35	7	4	0	0	10	10	10	25	30	35	40	45	45

Row 1 Item A: $w=2, p=10$

$$dp[1][2] = \max(dp[0][2], dp[0][0] + 10) = \max(0, 10) = 10$$

$$dp[1][w] = 10 \text{ for all } w \geq 2$$

Row 2 Item B: $w=5, p=25$

$$dp[2][5] = \max(dp[1][5], dp[1][0] + 25) = \max(10, 25) = 25$$

$$dp[2][7] = \max(dp[1][7], dp[1][2] + 25) = \max(10, 35) = 35$$

$$dp[2][10] = \max(dp[1][10], dp[1][5] + 25) = \max(10, 35) = 35$$

Row 3 Item C: $w=6, p=30$

$$dp[3][6] = \max(dp[2][6], dp[2][0] + 30) = \max(25, 30) = 30$$

$$dp[3][8] = \max(dp[2][8], dp[2][2] + 30) = \max(35, 40) = 40$$

$$dp[3][10] = \max(dp[2][10], dp[2][4] + 30) = \max(35, 40) = 40$$

Row 4 Item D: $w=7, p=35$

$$dp[4][7] = \max(dp[3][7], dp[3][0] + 35) = \max(35, 35) = 35$$

$$dp[4][9] = \max(dp[3][9], dp[3][2] + 35) = \max(40, 45) = 45$$

$$dp[4][10] = \max(dp[3][10], dp[3][3] + 35) = 45$$

Selected Items

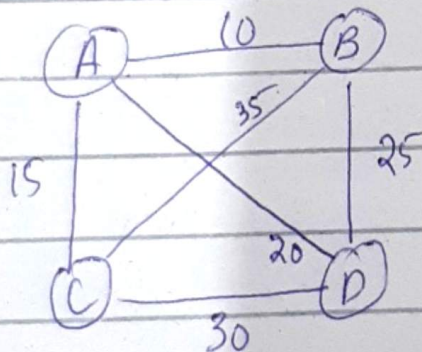
A B C D
 3-2=1 ← 10-7=3
 profit = 10 Profit = 35
 1 0 0 1

$$\text{Max profit} = 1 \times 10 + 0 \times 25 + 0 \times 30 + 1 \times 35 = 45$$

Time complexity = $O(NW)$

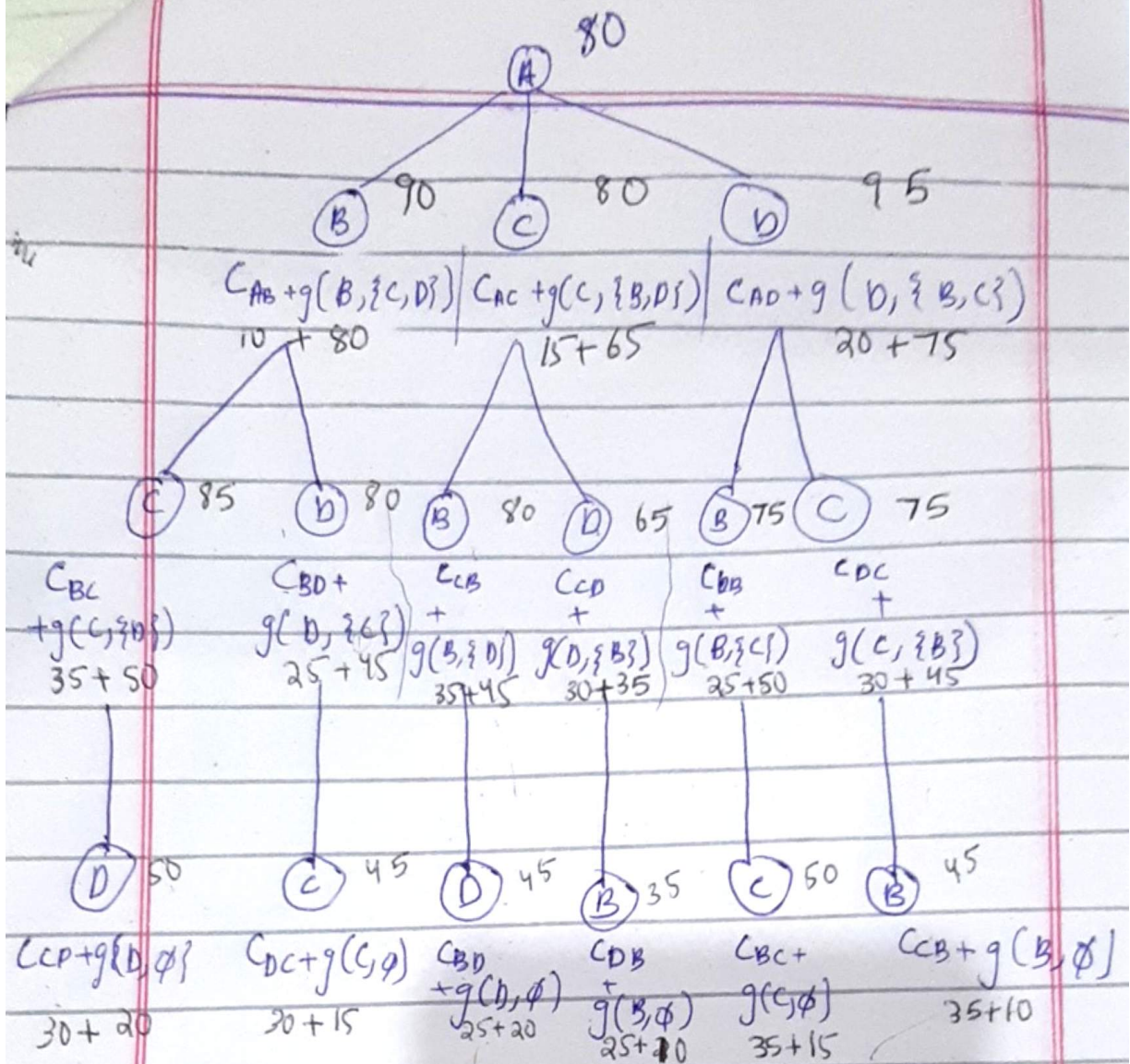
Space complexity = $O(NW)$

Q3



Starting At (A).

	A	B	C	D
A	0	10	15	20
B	10	0	35	25
C	15	35	0	30
D	20	25	30	0



$g(B, \emptyset) = 10$
 $g(C, \emptyset) = 15$
 $g(D, \emptyset) = 20$
 $g(C, \{D\}) = 50$
 $g(D, \{C\}) = 45$
 $g(B, \{D\}) = 45$
 $g(D, \{B\}) = 35$
 $g(B, \{C\}) = 50$
 $g(C, \{B\}) = 45$

Level 0:

Level 1:

Level 2:

$g(B, \{C, D\}) = 80$
 $g(C, \{B, D\}) = 65$
 $g(D, \{B, C\}) = 75$

$g(A, \{B, C, D\}) = 80$

Minimum Tour cost = 80

A → B → D → C → A

Complexity of Brute force Approach:

Time complexity = $O(N!)$

Space complexity = $O(N)$

Q4

The dp solution uses subsets of visited cities.

State Definition: let $DP(S, i)$

be the min cost to start from city A, visit all cities in subset S, & end at city i.

Where:

$S =$ set of visited cities containing A and i.

$i \in S$

Base case: When only the start city is visited,

$$DP(\{A\}, A) = 0$$

Transition: To reach city i, come from some city j already in the subset;

$$DP(S, i) = \min_{j \in S, j \neq i} \{ DP(S - \{i\}, j) + d(j, i) \}$$

where, $d(j, i) =$ distance from city j to city i.

After visiting all cities V, return to the starting city A:

$$\min_{i \neq A} \{ DP(V, i) + d(i, A) \}$$

where V is the set of all cities.

Time Complexity:

No. of subsets = 2^n

for each subset n possible ending cities
for each state, up to n transitions.

$$O(n^2 2^n)$$

Space Complexity:

$$O(n 2^n)$$

Q5

X = ABCBDAB

Y = BDCABA

		B	D	C	A	B	A
0		0	0	0	0	0	0
1	A	0	0	0	1	1	1
2	B	0	1	1	1	2	2
3	C	0	1	1	2	2	2
4	B	0	1	1	2	3	3
5	D	0	1	2	2	3	3
6	A	0	1	2	2	3	4
7	B	0	1	2	2	3	4

LCS: BCBA

~~BDAB~~

~~BCBA~~

Q6

		B	D	C	A	B	A
0		0	0	0	0	0	0
1	A	0	0	0	1	1	1
2	B	0	1	1	1	2	2
3	C	0	1	1	2	2	2
4	B	0	1	1	2	2	3
5	D	0	1	2	2	2	3
6	A	0	1	2	2	3	4
7	B	0	1	2	2	3	4

Time & space complexity = $O(n^2)$

LCS: BCBA

Q6

X = 10010101
Y = 01011010

		0	1	0	1	1	0	1	0	
		0	1	2	3	4	5	6	7	8
	0	0	0	0	0	0	0	0	0	0
1	1	0	0	1	1	1	1	1	1	1
0	2	0	1	1	2	2	2	2	2	2
0	3	0	1	1	2	2	2	3	3	3
1	4	0	1	2	2	3	3	3	4	4
0	5	0	1	2	3	3	3	4	4	5
1	6	0	1	2	3	4	4	4	5	5
0	7	0	1	2	3	4	4	5	5	6
1	8	0	1	2	3	4	5	5	6	6
			1		0	1	0	1	0	

LCS : 101010

Recurrence Relation.

$c[i, j] = 0$ if $i = 0$ or $j = 0$

$c[i, j] = c[i-1, j-1] + 1$ if $X[i] = Y[j]$

$c[i, j] = \max(c[i-1, j], c[i, j-1])$ if $X[i] \neq Y[j]$

Backtracking.

Starting at the bottom right $(8, 8)$ & tracing backward:

$(8, 8) \rightarrow$ Up to $(7, 8)$

$(7, 8) \rightarrow$ Match 0 $\rightarrow$ Diagonal to $(6, 7)$

$(6, 7) \rightarrow$ Match 1 $\rightarrow$ Diagonal to $(5, 6)$

$(5, 6) \rightarrow$ Match 0 $\rightarrow$ Diagonal to $(4, 5)$

$(4, 5) \rightarrow$ Match 1 $\rightarrow$ Diagonal to $(3, 4)$

$(3, 4) \rightarrow$ Up to $(3, 3)$

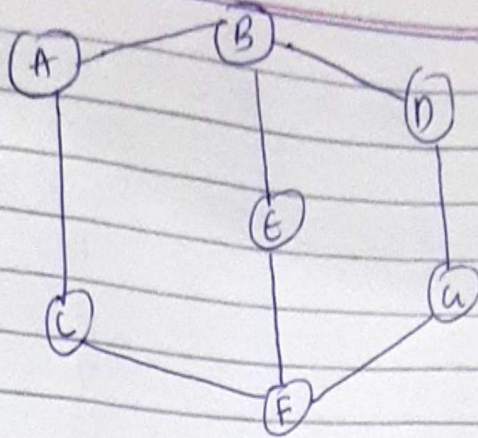
$(3, 3) \rightarrow$ Match 0 $\rightarrow$ Diagonal to $(2, 2)$

$(2, 2) \rightarrow$ Up to $(1, 2)$

$(1, 2) \rightarrow$ Match 1 $\rightarrow$ Done.

Reversing the matches gives the LCS: 101010

Q2



Adjacency List

A: B, C

B: A, D, E

C: A, F

D: B, G

E: B, F

F: C, E, G

G: D, F

Queue

[A]

Pop A [B, C]

Pop B [C, D, E]

Pop C [D, E, F]

Pop D [E, F, G]

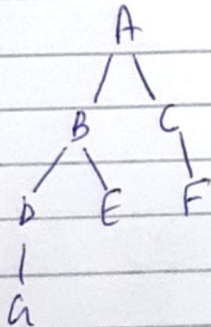
Pop E [F, G]

Pop F [G]

Pop G []

BFS Traversal Order

A → B → C → D → E → F → G



Shortest distance from A

A	0
B	1
C	1
D	2
E	2
F	2
G	3

Time complexity = $O(V+E)$

vertices
↓
Edges
visited

Space Complexity = $O(V)$

Q8

$A \rightarrow B \rightarrow D \rightarrow G \rightarrow F \rightarrow C \rightarrow E$

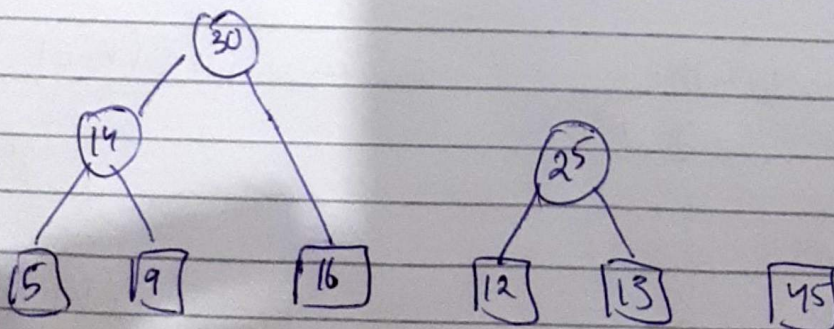
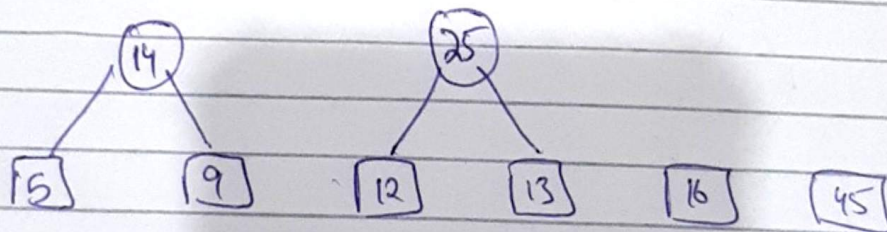
Node	Discovy time	Finish time
A	1	14
B	2	13
C	6	7
D	3	12
E	8	9
F	5	10
G	4	11

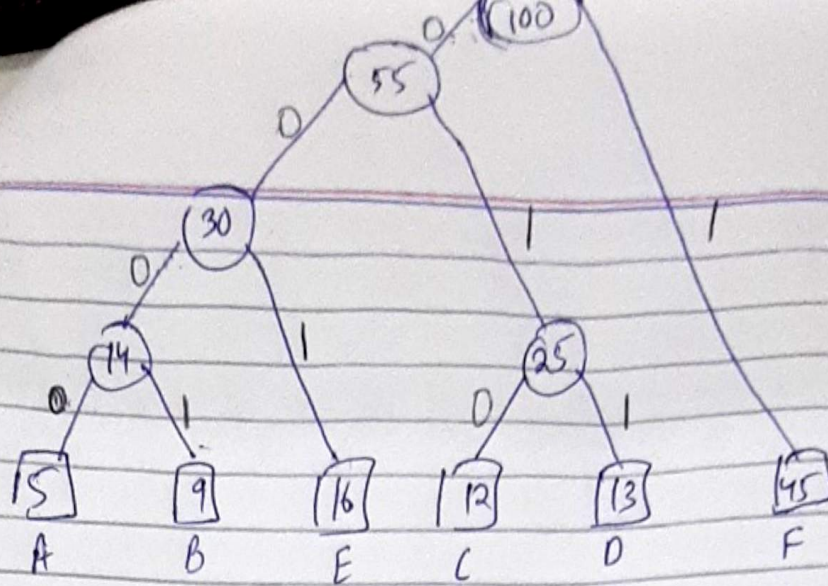
DFS Tree Edges

$(A, B), (B, D), (D, G), (G, F), (F, C), (F, E)$

Time & Space Complexity = $O(V+E)$

Q9





0000
4 bits

0001
4 bits

001
3 bits

010
3 bits

011
3 bits

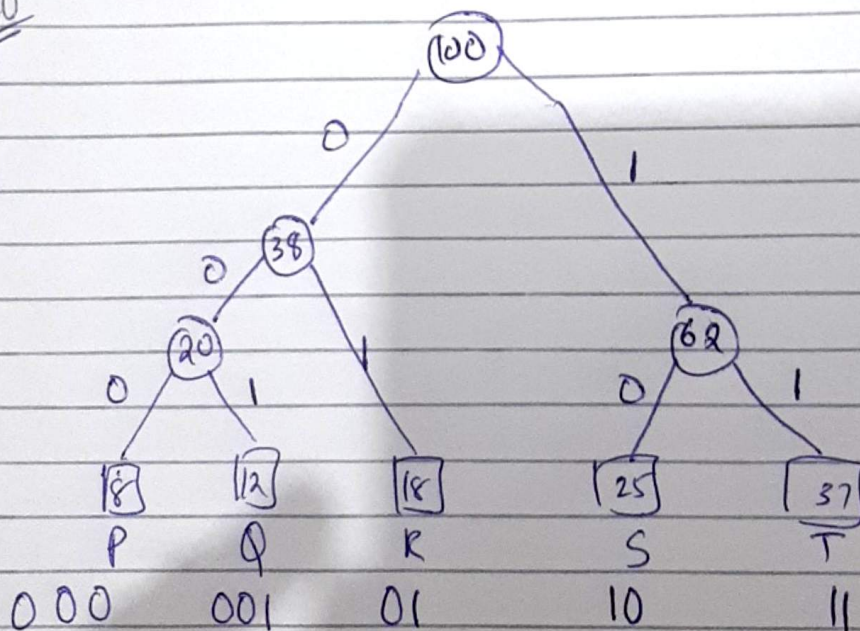
1
1 bit

$$5 \times 4 + 9 \times 4 + 16 \times 3 + 12 \times 3 + 13 \times 3 + 45 \times 1 = 224 \text{ bits}$$

Time Complexity $\rightarrow O(N \log N)$

Space $\rightarrow O(N)$

Q10



000

001

01

10

11

$$8 \times 3 + 12 \times 3 + 18 \times 2 + 25 \times 2 + 37 \times 2 = 220 \text{ bit}$$

$$\text{Avg} = \frac{220}{100} = 2.2 \text{ bit/char}$$

A Huffman code is guaranteed to be prefix free because all actual characters are stored exclusively at the leaf nodes of the constructed binary tree. Because no leaf node can be a parent or ancestor of another node, it is geometrically impossible for the path of one character to be a prefix of the path of another.

Time complexity $\approx O(N \log N)$, N dehniques
Space $\approx O(N)$